Adelaide University

Microcontroller Programming

Project

Autonomous Mobile Robot

NUCLEO-G474RE • Kobuki robot

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1. Project Overview

This project has four assessment components:

- Final fully commented code due Friday 6 November 2026, 23.59;
- Report and Presentation video due Friday 6 November 2026, 23.59;
- Robot demonstration which can be conducted during any practical session. There will be one final session during the extension week (Week 12) that we will be used to test your final algorithm.

You are expected to submit four files via the myLearning submission page by the due date of 23.59 Friday 6 November, 2026:

1. A 20-page report with a filename of 'group number day' in '.pdf' format. For example: 'group16 tue1pm' or'group05 wed2pm'. Clearly display all group members names and student ID's on a title page.
2. A folder titled 'group number day' in '.zip' format. This folder that contains all project files created by the VS Code. In addition, your project must be saved-as 'group number day' and within each '.c' and '.h' file you must write your student numbers as a comment on the first line.
3. A video presentation discusses the overall program and Finite State Machine used in the project, which clearly show the states descriptions and switching conditions. The presentation is 12 minute long and each member must speak for 4 minutes with clear and succinct slides.
4. A video showing your robot completing the two courses as described later in the document.

This is a group assignment with a maximum group size of three people; only one group member will need to submit. Each member will also need to complete a peer review at the end of the project. That is, you could receive a low grade if there is no evidence of contribution to the project. **A statement of contribution must be included at the beginning of your report, outlining what each member contributed to the project itself and the written report.** Your report will be checked for plagiarism using Turnitin. We have methods to check for plagiarism of the code (not inclusive of guided practical code).For further details on the marking scheme, please view the rubric on the myLearning page.

2. Introduction

You are tasked with programming an odometry and object-detection algorithm for an Autonomous Mobile Unit (AMU) for use at a parcel sorting company, similar to that shown in Figure 1. Your supervisor has asked you to program and test this using a Nucleo G474RE microcontroller and a Kobuki autonomous mobile robot programmed via VS Code. The USART port of the Nucleo G474RE is to be used to communicate with the

Kobuki robot via an asynchronous (UART) protocol. Further, you are provided with an ultrasonic sensor to assist with your task; this is either an analog sensor or a sensor which communicates via UART (the type of sensor you are given will be explicitly told to you). You will provide a project specification report and an in-person demonstration of the AMU operation to your supervisor upon completion.



Figure 1: Small delivery robot at a parcel sorting station. Source:Supply Chain IT

Parcels from a sorting station need to be transported to a nearby storage unit. To minimise the involvements of humans, these are to be moved and stored using an AMU once a package has been placed on top of the robot. In order to avoid any humans working in the corridors, the AMR has two available paths to the storage unit. The humans kindly place bollards to guide the AMU toward the correct path; however, if the AMU were to drive into a bollard, this may cause significant damage to the robot and its package - thus this must be avoided.

Your microcontroller will sit in an idle state until receiving a start signal from the Kobuki microcontroller. Upon receiving a start signal, the microcontroller will send a message instructing the AMU to scan the surroundings, then navigate along one of two paths to the storage unit, before returning to an idle state whilst the package is loaded into the deposit location (your boss has forgotten about the return trip, but it is best to not remind him). Due to limitations in the storage space, precise alignment is required by the AMU to interact with the deposit location. The AMU should also go back to idle if it receives a stop signal at any point from the Kobuki microcontroller (i.e. pressing of the button B1 on the Kobuki).

For a satisfactory outcome, any trip to the storage unit must be completed within a maximum specified period of time. The AMU must navigate with assistance of an ultrasonic sensor to locate both the bollards, and to align with the deposit location.

The AMU also has multiple bumper switch sensors in order to stop precisely, or as a fail-safe in-case an obstruction was detected. The warehouse is surrounded by a drainage ledge, so to prevent driving into this, multiple LiDAR sensors are fitted in the same

locations as the bump switch sensors. For safety, the AMU should stop when a button is pressed (you can choose which button) or the robot is lifted off the ground.

3. Test Area Specification

Figure 2 contains all dimensions (in mm) and references for the test area. The AMU will always begin in the same package collection location, but at an arbitrary orientation. There are four possible path combinations: two bollards in either the A or B positions, blocking both ends of one path to the storage unit; one storage deposit location, in one of two locations. The Kobuki AMU will be tested on two unknown layouts, and must complete both back-to-back, using the same program, a simplified version of the testing layouts are shown in Figure 3. Success will be dependent upon the time it takes to deposit the package in the storage and whether the centre of the robot face the deposit spot.

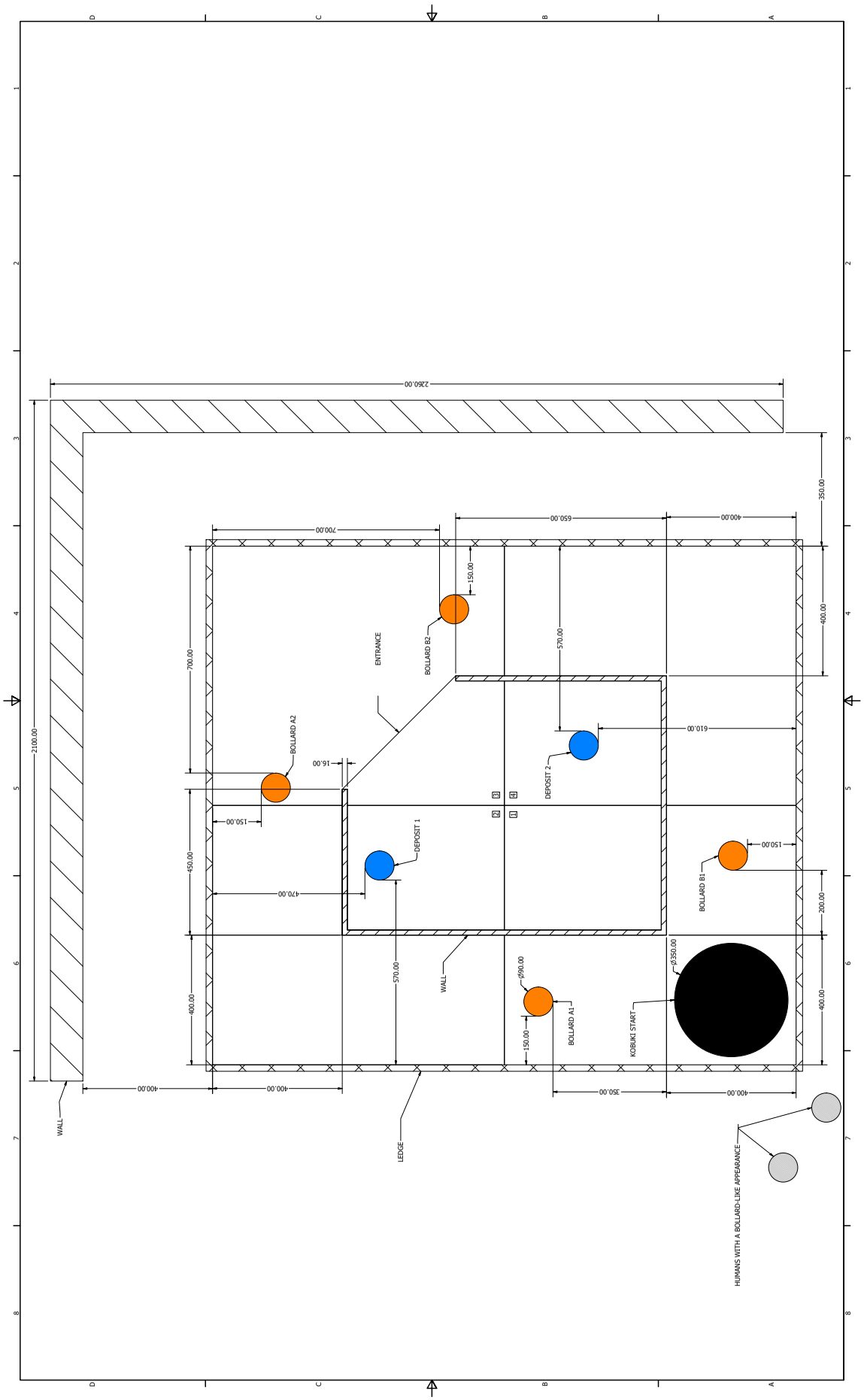


Figure 2: Test Area Specification

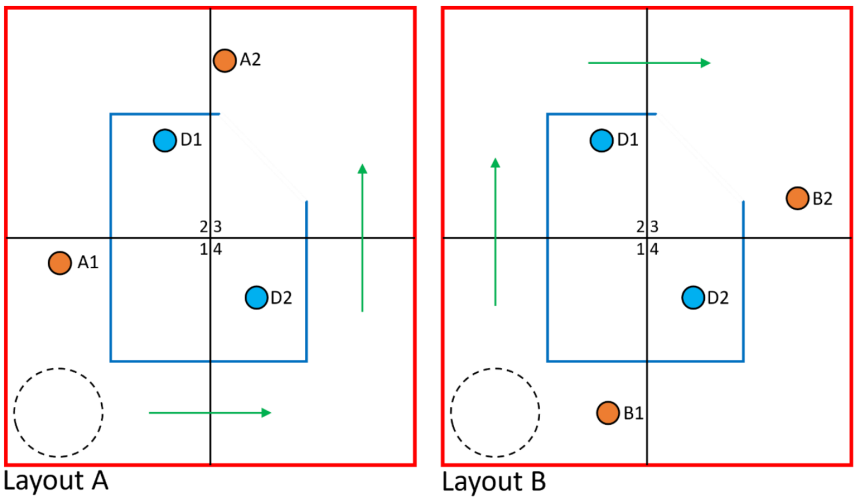


Figure 3: The track has two bollards arranged in either layout A or B, and the deposit may be in one of two locations D1 or D2. The specific layout and deposit location will not be known prior to testing. Hence, the AMR should be programmed so that it can identify and successfully navigate whichever layout it is on and have the ability to find the deposit at whichever location it is placed at. Additionally, the AMR should be able to start from any orientation (i.e., facing any arbitrary direction) and still be able to orientate itself and function as specified.

Bollards (representing human presence) are 90mm in diameter, shown by orange circles. The deposit location is marked with a bollard-sized marker, shown by blue circles. The Kobuki starting area is shown by a black circle in Figure 2 and dotted circle in Figure 3. There are two fixed walls bounding half of the rectangular area; however, the other half may have erroneous humans, which is shown by two external bollards randomly placed closed to the platform (as described by the two grey circle outside of the platform). You can use these two bollard to help orientate your robot.

Throughout the practical sessions, you can achieve progression marks as described by Table 1. You can have your robot tested as many time as you like within the practical sessions. In the last session, your AMU will be tested back-to-back on each layout with the same program running (you will not be able to reprogram between each layout as you are with tasks described in Table 1). You will not be able to touch your robot between each layout, marks are awarded as shown in Table 2.

Safety checks will be conducted to ensure the surrounding crowd are not about to be attacked by a battlebot. That is, the robot must be responsive to a stop button and wheel drop. Throughout the run, **your robot must never exceed the 150 mm/s speed limit.**

Table 1: Progression goals - marks awarded in any practical session.

Task	Weighting, %
Layout A - pass safety checks	0.5
Layout A - enter section 4 within **30** seconds	1
Layout A - enter section 3 within **60** seconds	1
Layout A - entire Kobuki within the storage unit	2
Layout A - avoided hitting any bollard	1
Layout A - stop 100 mm away from the deposit location 2	1
Layout A - stop aligned with deposit location 2	1
Layout B - pass safety checks	0.5
Layout B - enter section 2 within **30** seconds	1
Layout B - enter section 3 within **60** seconds	1
Layout B - entire Kobuki within the storage unit	2
Layout B - avoided hitting a bollard	1
Layout B - stop 100mm away from the deposit location 1	1
Layout B - stop aligned with deposit location 1	1
	15

Table 2: Final algorithm test - one program - to be performed by tutors in the last session. Layout 1 and 2 are randomly determined by the location of the orange and blue bollards

Task	Weighting, %
Layout 1 - pass safety checks	1
Layout 1 - enter section 2 or 4	0.5
Layout 1 - enter section 3	0.5
Layout 1 - entire Kobuki within the storage unit	1
Layout 1 - avoided hitting a bollard	1
Layout 1 - stop 100mm away from the deposit location 2	2
Layout 1 - stop aligned with deposit location 2	2
Layout 2 - enter section 2 or 4	0.5
Layout 2 - enter section 3	0.5
Layout 2 - entire Kobuki within the storage unit	1
Layout 2 - avoided hitting a bollard	1
Layout 2 - stop 100mm away from the deposit location 1	2
Layout 2 - stop aligned with deposit location 1	2
	15

4. Assessments

1. **Report**: The report does not have a mandated structure, but will be expected to contain: a project description, your program design choices, the algorithm for the robot path and object detection, a review of the final functionality, flowcharts and finite state machine diagram(s), and a description of any unique functionality included that was not given within the practicals. The report has a 20-page limit, excluding the appendices. - 30%

2. **Code**: Your submitted project folder must contain all wrappers, header files and subroutines/source code. The majority of wrapper functions and subroutines have been provided for you during the practicals. Your primary coding assessment pertains to your ability to devise, construct and implement a Finite State Machine which goes beyond what you have been provided to complete the project (i.e. practical 5 and 6 content). That is, you are encouraged to introduce new wrappers and subroutines as you see fit to meet the provided specification. Please fully comment your code files, and document the novel code within your report. Your code project will be built to check for errors. - 20%

3. **Presentation**: A video presentation describing how your AMU completed each layout and the functionality you implemented to achieve this. Within the presentation, please discuss your overall program development, code prototyping (issues encountered in development and how you remedied in final algorithm) and detailed discussion on the Finite State Machine you implemented. The presentation is 12 minutes long and involves all group members, i.e. 4 minutes each. - 20%

4. **Functionality**: Robot functionality demonstrated in the project practical sessions over Weeks 9, 10 and 11; and the final algorithm in week 12. This can only be done in person with a tutor marking your group off; and is based on the progression marks tabulated in Tables 1-2. - 30%